

Unit Rates and Constant Speed

USA Grade 6 Math
CCSS 6.RP.A.2-3

Name: _____

Date: _____

LEARNING TARGET

I can calculate and compare unit rates in real-world situations.

KEY STRATEGY + WORKED EXAMPLE

Divide both quantities by the second quantity to find an amount per 1, include units, and compare like units.

210 miles in 3 hours = 70 miles per hour.

A. TRY TOGETHER - USE THE STRATEGY

- 180 units in 3 hours: unit rate = ____ per hour
- 245 units in 5 hours: unit rate = ____ per hour
- 96 units in 4 hours: unit rate = ____ per hour
- 525 units in 7 hours: unit rate = ____ per hour

B. CORE PRACTICE - SHOW YOUR WORK

5. 144 units in 6 hours: unit rate = ____ per hour

6. 360 units in 8 hours: unit rate = ____ per hour

7. 275 units in 11 hours: unit rate = ____ per hour

8. 420 units in 12 hours: unit rate = ____ per hour

9. 156 units in 3 hours: unit rate = ____ per hour

10. 672 units in 14 hours: unit rate = ____ per hour

11. 180 units in 3 hours: unit rate = ____ per hour

12. 245 units in 5 hours: unit rate = ____ per hour

C. INDEPENDENT FLUENCY - CALCULATE AND CHECK

13. 96 units in 4 hours: unit rate = ____ per hour

14. 525 units in 7 hours: unit rate = ____ per hour

15. 144 units in 6 hours: unit rate = ____ per hour

16. 360 units in 8 hours: unit rate = ____ per hour

17. 275 units in 11 hours: unit rate = ____ per hour

18. 420 units in 12 hours: unit rate = ____ per hour

19. 156 units in 3 hours: unit rate = ____ per hour

20. 672 units in 14 hours: unit rate = ____ per hour

Unit Rates and Constant Speed

USA Grade 6 Math
CCSS 6.RP.A.2-3

Name: _____

Date: _____

D. APPLY THE SKILL

21. Plan A costs \$24 for 6 GB; Plan B costs \$35 for 10 GB. Which has lower cost per GB?

22. A cyclist travels 42 miles in 2.5 hours. Find average speed.

23. A faucet fills 18 gallons in 7.5 minutes. How much in 20 minutes at the same rate?

E. REASON, CHECK AND CORRECT

24. Why must units match before comparing two rates?

25. Correct: 150 miles in 2 hours is 152 mph.

EXIT TICKET + CHALLENGE

26. Find and interpret the unit rate for \$18.75 per 6 pounds.

Challenge: Create two competing-price offers and determine the better buy.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: _____

31. The most difficult part was: _____

Why was it difficult? _____

32. My next practice step is: _____

TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

- A. 1) 60, 2) 49, 3) 24, 4) 75
B. 5) 24, 6) 45, 7) 25, 8) 35, 9) 52, 10) 48, 11) 60, 12) 49
C. 13) 24, 14) 75, 15) 24, 16) 45, 17) 25, 18) 35, 19) 52, 20) 48
D. 21) Plan B: \$3.50/GB versus \$4/GB. 22) 16.8 mph. 23) 48 gallons.
E. 24) Unlike units do not measure the same comparison. 25) $150/2=75$ mph.
EXIT. 26) \$3.125 per pound (about \$3.13).

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may add quantities or omit units. A unit rate is a quotient with a denominator of one.

TEACHER CHECK

- ☐ Uses an appropriate Grade 6 Math strategy.
- ☐ Shows or explains thinking clearly.
- ☐ Checks a response using evidence, a rule, or a second method.
- ☐ Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.